Joint Cost–Performance Optimization of CDN Disk Capacity

1. Setting

A content-delivery network (CDN) serves traffic to end-users through a global set of **edge metros** — points of presence (PoPs) each identified by an airport code (e.g., `BOS`, `LHR`, `SIN`). Each metro runs one or more **edge caches**. When a request arrives at an edge cache, it is either:

- **a hit** — the content is already on disk and is served immediately, or
- **a miss** — the edge cache must fetch the content from a **mid-tier cache hub (MCH)**, incurring extra round-trip time and bandwidth cost.

Each metro belongs to a **geo** (e.g., North America, EMEA, APAC) and may serve traffic originating from many neighboring metros. The question we answer is:

> **How much disk should be provisioned at each edge metro to jointly minimize infrastructure cost and end-user latency, given the available content footprint at each site?**

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2. Inputs

2.1 Footprint Descriptor (FDS)

For each metro `m` we have a **footprint descriptor** — an empirically measured curve:

$$h_m : s \mapsto h_m(s) \in [0, 100]$$

where `s` is the cache disk allocation (in MB) and `h_m(s)` is the resulting cache **hit-rate** (%). This curve is monotonically non-decreasing and saturates at a metro-specific maximum (e.g., Boston's FDS saturates at ~77%). The FDS captures the working-set structure of the traffic mix served at that metro.

2.2 Traffic

A **traffic matrix** `T_{ij}` (Mbps) records how much traffic from end-user metro `i` is served by edge metro `j`. From this we derive:

- **Incoming traffic** `\lambda_j = \sum_i T_{ij}`: total Mbps flowing into edge metro `j`.
- **Neighborhoods**: metro `j` is an active serving neighbor of `i` if `T_{ij}` exceeds a configurable threshold (e.g., 5 Gbps). Only these edges are considered during optimization.

2.3 Latency Distributions

Latency at each hop is modeled as a **probability density function (PDF)** over milliseconds:

Component	Notation	Meaning
Edge RTT	f^{rtt}_{ij}	Round-trip time from user metro i to edge metro j
Edge TAT (hit)	$f^{\text{tat-hit}}_j$	Turn-around time at edge on a cache hit
Edge TAT (miss)	$f^{\text{tat-miss}}_j$	Turn-around time at edge on a cache miss (includes MCH fetch)

PDFs are either loaded from empirical measurements or approximated as Gaussians. Convolution of independent latency components is performed via FFT.

2.4 Cost Model

The monthly cost of operating edge metro j with disk allocation s_j and incoming traffic λ_j is decomposed into:

$$C_j(s_j, \lambda_j) = r_j \cdot \left[C^{\text{dep}}(s_j) + C^{\text{colo}}(s_j) + C^{\text{mid}}(\lambda_j, h_j(s_j)) + C^{\text{parent}}(\lambda_j, h_j(s_j))\right]$$

where:

- $r_j \in \{2, 3, 5\}$ is the **replication factor** (determined by metro tier: Tier-2 = 2×, Tier-1 = 3×, Tier-0 = 5×),
- C^{dep} is disk **depreciation** cost (capital amortization),
- C^{colo} is **colocation** cost (rack space, power),
- C^{mid} is **midgress** cost (bandwidth from edge to MCH on misses),
- C^{parent} is **parent service** cost (MCH serving the miss traffic).

Traffic-proportional costs ($C^{\text{mid}}, C^{\text{parent}}$) scale with the miss fraction $(1 - h_j(s_j)/100)$ and are computed on the per-replica split traffic λ_j / r_j.

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3. Performance Model

3.1 Time-to-First-Byte (TTFB)

For a user at metro i served by edge metro j with hit-rate h, the TTFB is a random variable whose distribution is:

$$\mathrm{TTFB}_{ij}(h) = f_{ij}^{\text{rtt}} \;\circledast\; \left[\frac{h}{100} \cdot f_j^{\text{tat-hit}} + \left(1 - \frac{h}{100}\right) \cdot f_j^{\text{tat-miss}}\right]$$

where $\circledast$ denotes convolution. In words: every request pays the RTT; then it additionally pays either the hit TAT or the miss TAT according to the current hit-rate.

3.2 Aggregate Performance per Metro

Traffic from multiple user metros is blended at edge metro j:

$$F_j = \sum_i \frac{T_{ij}}{\sum_k T_{kj}} \cdot \mathrm{TTFB}_{ij}(h_j)$$

The **P50** and **P95** of F_j are the key performance metrics, denoted τ^{50}_j and τ^{95}_j respectively.

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4. Objective Function

We seek a disk allocation $\mathbf{s} = (s_1, \ldots, s_M)$ for all M active metros that minimizes the combined cost and latency penalty:

$$\boxed{\min_{\mathbf{s}} \;\; \mathcal{L}(\mathbf{s}) = \underbrace{\sum_{j=1}^{M} C_j\!\left(s_j,\, \lambda_j\right)}_{\text{Infrastructure cost}} + \underbrace{\sum_{j=1}^{M} \Psi_j\!\left(\tau^{50}_j(\mathbf{s}),\, \tau^{95}_j(\mathbf{s})\right)}_{\text{Latency penalty}}}$$

4.1 Penalty Function

The latency penalty for metro j is:

$$\Psi_j(\tau_j^{50},\, \tau_j^{95}) = 2\,\lambda_j^{\text{GB}} \cdot \max\!\left(\tau_j^{50} - \bar{\tau}_j^{50},\, 0\right)^2 + 2\,\lambda_j^{\text{GB}} \cdot \max\!\left(\tau_j^{95} - \bar{\tau}_j^{95},\, 0\right)$$

where:

- $\lambda_j^{\text{GB}} = \lambda_j / 1000$ is the outbound traffic in Gbps (used as a traffic-weighted importance scalar),
- $\bar{\tau}^{50}_j,\; \bar{\tau}^{95}_j$ are **regional latency targets** for metro j.

Asymmetry by design: the P50 penalty is **quadratic** in the excess — small violations near the target cost little, but large deviations are penalized aggressively. The P95 penalty is **linear** — every millisecond of tail-latency excess costs proportionally regardless of magnitude. This reflects the engineering intuition that P95 tail control is non-negotiable while P50 has diminishing marginal value once already near target.

4.2 Regional Targets

Latency targets differ by geographic region, reflecting realistic network physics:

Region	$\bar{\tau}^{50}$ **(ms)**	$\bar{\tau}^{95}$ **(ms)**
Europe	24	105
Middle East	45	180
Africa	75	220
South Africa	35	200
North America (default)	24	105

4.3 Coupling Between Metros

The objective is **not separable** across metros. The disk allocation s_j at metro j affects $h_j(s_j)$, which changes τ^{50}_j and τ^{95}_j, which in turn affects Ψ_j. However, Ψ_j is itself weighted by the traffic arriving from *all neighboring user metros*, so the gradient of the penalty at j depends on the full neighborhood traffic structure. This coupling necessitates a global optimization procedure.

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5. Optimization Algorithm

5.1 Initialization

Each metro starts at its **cost-optimal point** — the disk allocation s_j^* that minimizes C_j alone (ignoring latency). This is found by scanning the FDS curve and evaluating cost at each integer hit-rate percentile.

5.2 Gradient Computation

At each iteration, the **marginal gradient** for metro j is computed by probing a fixed disk increment Δs (5 TB):

$$g_j = \left[C_j(s_j + \Delta s) - C_j(s_j)\right]_{\text{cost}} + \left[\Psi_j(s_j + \Delta s) - \Psi_j(s_j)\right]_{\text{perf}}$$

A **negative gradient** ($g_j < 0$) means adding disk reduces the combined objective — i.e., the latency savings outweigh the infrastructure cost increase.

Gradients are computed for all metros **in parallel** using a thread pool.

5.3 Budget-Proportional Update

Rather than a fixed step size, a **budget** is allocated each iteration:

$$B = N^{-}_{\text{iter}} \times \Delta_{\text{TB}}$$

where N^{-}_{iter} is the number of metros with a negative gradient and $\Delta_{\text{TB}} = 10\,\text{TB}$ is the per-metro budget quantum. The budget is distributed across metros proportionally to the magnitude of their gradient:

$$\delta s_j = \text{round\_up\_TB}\left(\frac{|g_j|}{\sum_k |g_k|} \cdot B \right)$$

Metros with $g_j < 0$ receive additional disk; metros with $g_j > 0$ lose disk (down to their FDS minimum). This implements a soft form of **Frank-Wolfe / conditional gradient** descent adapted to the discrete FDS structure.

5.4 Convergence

The process iterates until the objective $\mathcal{L}$ stops improving. At each iteration the per-metro state (disk, hit-rate, cost, penalty, P50, P95, gradient components) and the summary-level objective are logged to disk for post-hoc analysis.

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6. Summary of Structural Assumptions

Assumption	Justification
FDS curve is fixed and pre-measured	Cache hit-rate depends on content working set, not on optimizer decisions made here

Assumption	Justification
Latency distributions are independent across hops	Standard queueing approximation; validated empirically
Cost model is separable across metros	Each PoP is billed independently
Replication factor is fixed per metro tier	Determined by reliability SLAs, not by cost optimization
Traffic matrix is stationary	Optimization runs on a representative traffic snapshot; re-run periodically as traffic shifts
P50 penalty quadratic, P95 penalty linear	Engineering policy choice reflecting tail-latency importance